Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/01/2006
PI Name:	Singh, R Paul Ph.D.		
Project Title:	A Multipurpose Fruit and Vegetable Processing System for Advanced Life Support		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	01/13/2003	**End Date:**	01/12/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The following tasks were identified in the proposal: 1) Year 1 (January 1, 2003-December 31, 2003): a. Select varieties of tomatoes and other vegetables suitable for processing b. Develop conceptual design for the MFVP c. Develop computer-aided simulations of each operation to assist in design d. Build bench-scale components of the MFVP and conduct preliminary trials for technical feasibility e. Select optimum design parameters f. Select membranes for water cleaning and product recovery 2) Year 2 (January 1, 2004 – December 31, 2004) a. Construct MFVP from components b. Select membrane systems for concentration of tomato juice c. Conduct trials with MFVP to determine operational feasibility d. Determine ESM and other operational variables e. Conduct optimization trials 3) Year 3 (January 1, 2005 – December 31, 2005) a. Conduct experimental trials to determine safety, quality and		

	acceptability of foods processed in MFVP b. Develop final modifications as indicated from the experimental data c. Write final project report To develop a miniaturized fruit and vegetable processor and to evaluate and optimize its performance for ALS The development of the fruit and vegetable processor represents the first attempt to miniaturize and integrate the array of unit operations required for preparation and processing of fruits and vegetables such as tomato. Additionally, the developers integrate the ESM metric evaluation throughout the development process as the primary project management tool.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research data on concentrating tomato juice using new generation membranes is valuable for the food industry. Only one Japanese company is currently employing this technique in a food plant located in California. Our data should prove useful for those food processors who are seeking optimization of the process.
Task Progress:	Based on the results obtained from the second year of the project (reported in the last annual report) we constructed a second generation prototype of the multipurpose fruit and vegetable processor (MFVP). While some of the components from the first generation MFVP were used, a number of new parts were custom designed and fabricated. The second generation processor operates in a continuous mode to produce sliced tomatoes, diced tomatoes, tomato juice, tomato sauce, and tomato paste from raw tomatoes. In addition to constructing the second generation MFVP, additional trials were carried out with the reverse osmosis membrane system for concentration of tomato juice. ESM values for a reverse osmosis system and the complete MFVP system were determined. Future trials, planned for the fourth year (no-cost extension), include studying a novel system for cleaning-in-place of the MFVP using electrolyzed water. Briefly, some of the key results with the membrane system were as follows: the polyacrylamide membranes, used in the reverse osmosis of tomato juice, were nearly 100% effective in separating solids. The total solids increased asymptotically, and approached a maximum value as the permeate flux decayed. Tomato juice with an initial value of 7.0 ±0.1 % solids [w/w] was concentrated to 18.9 ±0.7 % solids [w/w] in 47 minutes. Rejection values for soluble solids appeared to increase with concentration. Electrical conductivity measurements were used to evaluate the leakage of small soluble electrolytes across the membrane. It was found that initially, rejections were high averaging greater than 99%, toward the end of concentration the value dropped to 98%. Reduction in processing times by operating at higher temperatures and pressure reduces the total energy of the concentration cycle. In the second generation MFVP, the size reduction device was operated in the vertical operation to allow gravity feed as low gravity will be present on the Mars surface. The hot break procedure was performed using direct steam injection. A steam venturi acts as a stationary pump to agitate and heat the tomato dices. Vacuum transfer was selected to transport the tomatoes through the finisher screen. During the construction of the second generation MFVP, considerable delay in obtaining stainless steel was experienced due to shortages. In addition, the overall design required more time to fabricate the unit. Due to these unexpected delays, a one year no-cost extension to January 2007 was requested. During 2006, studies will be conducted to use a novel system for cleaning the MFVP. A prototype pipe transport system will be constructed. Electrolyzed water will be used to clean and disinfect the system. The performance of this new cleaning system will be evaluated. A final project report will be prepared to document results obtained during the total duration of the project.
Bibliography Type:	Description: (Last Updated: 03/06/2007)
Articles in Peer-reviewed Journals	Voit DC, Santos MR, and Singh RP. "Development of a multipurpose fruit and vegetable processor for a manned mission to Mars." Journal of Food Engineering, in press, July 2006. , Jul-2006